ES · ITPM · ASTMA

Smart Book Review Management & Customer Opinion Analytics

Theory behind the system: text mining, enterprise systems and IT project management

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One system, three subjects

The project turns raw customer reviews into business decisions by combining three coursework areas.

1 **ASTMA**

Applied Statistics & Text Mining. Preprocessing, sentiment analysis and opinion mining on book review text.

2 **ES**

Enterprise Systems. Database, book catalog and review management, and the analytics dashboard.

3 **ITPM**

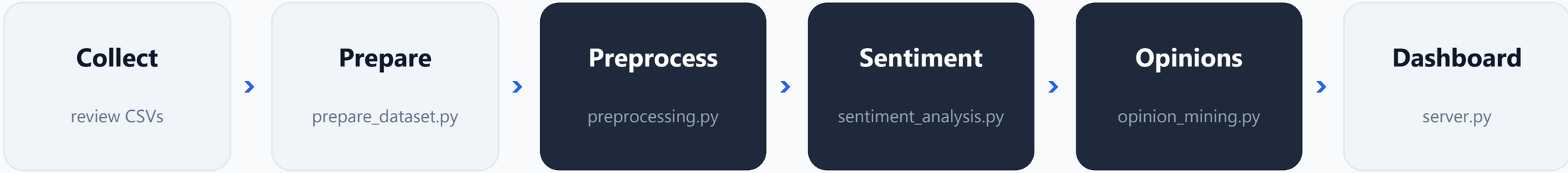
IT Project Management. WBS, schedule, resources, cost and risk planning.

124+ reviews analysed

3 sentiment classes

10 opinion aspects

The text-mining pipeline



Dark boxes are the ASTMA stages.

Why this order?

Each stage feeds the next. Clean, uniform data gives more reliable scores, and scores plus aspects power the dashboard KPIs.

Text preprocessing

1. **Lowercasing**
   Masterpiece and masterpiece become one token

2. **Noise removal**
   Strip URLs, numbers, punctuation symbols

3. **Tokenisation**
   Split review text into word tokens

4. **Stopword removal**
   Drop the, is, and, but

5. **Lemmatisation**
   reading becomes read

BEFORE

The novel has amazing character development, but the pacing made the middle chapters very boring!

AFTER

novel amazing character development pacing make middle chapter bore

Three ways to classify sentiment

Lexicon-based

Scores words against a pre-rated dictionary.

Strength: Fast, no training data, explainable

Limit: Weak on sarcasm and context

Used here: VADER

Machine learning

Trains Naive Bayes or SVM on labelled reviews.

Strength: Learns domain wording

Limit: Needs labelled data

Deep learning

Transformers such as BERT learn context.

Strength: Best accuracy on nuance

Limit: Heavy compute and data

How VADER scores a review

1. **Punctuation** !!! raises intensity
2. **Capitalisation** GREAT is stronger than great
3. **Degree modifiers** very, slightly
4. **Negation** not good flips polarity
5. **Contrast** but shifts weight to the end

Compound score: -1 to +1

Negative	Neutral	Positive
<= -0.05	in between	>= +0.05

The output also includes pos, neu and neg proportions. Your code classifies using only the compound value.

Opinion mining: what is the review about?

A keyword dictionary maps each aspect to indicator words, matched with word-boundary regex. Simple, fast and explainable.

- plot
- characters
- pacing
- writing style
- ending
- world building
- emotion
- theme
- cover design
- price

REVIEW

The novel has amazing character development but the pacing was slow and the ending was rushed.

ASPECTS DETECTED

characters, pacing, ending

Enterprise system architecture

Presentation layer

server.py & static web dashboard (HTML/CSS/JS)

Business logic layer

sentiment_analysis.py, opinion_mining.py

Data layer

SQLite: books, users and reviews tables

What makes it an enterprise system

- Integrates data and processes on one shared platform
- Supports CRUD on books and reader reviews
- Turns stored data into decision support

Data layer and business intelligence

1 Relational database

Books and reviews tables with CRUD operations in SQLite.

2 Indexing

Indexes on sentiment, genre, rating, and book title speed up filters.

3 Chunked processing

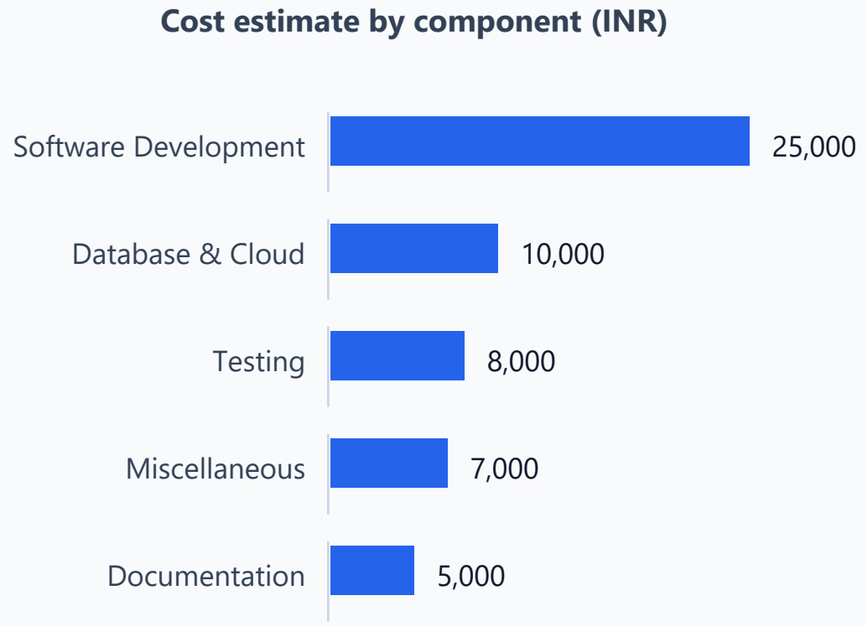
Reviews load in batches so entire corpus fits in memory efficiently.

4 BI dashboard

KPIs, distributions and aspect charts, cached for 5 minutes.

IT project management

1. **WBS** Break the project into work packages
2. **Schedule** 16 weeks with a Gantt-style view
3. **Milestones** Eight checkpoints, week 1 to 16
4. **Resources** Roles allocated by percentage
5. **Cost** Bottom-up budget estimate
6. **Risk** Rate, then mitigate



Rs. 55,000 total estimated cost

Risk register

Risk	Probability	Impact	Mitigation
Poor review data quality	Medium	High	Clean and validate review data
Incorrect sentiment classification	Medium	Medium	Test NLP model with sample reviews
Database failure	Low	High	Maintain database backup
Project schedule delay	Medium	High	Track schedule and milestones
Integration problems	Medium	Medium	Test modules independently first
Insufficient testing	Low	Medium	Prepare test cases

Limitations to acknowledge

1 Preprocessing clashes with VADER

Removing stopwords and punctuation strips negation and emphasis cues.

2 Document-level sentiment

One score per review, so aspects are detected but not scored separately.

3 Keyword matching

Misses synonyms and context; a word like good fits many aspects.

4 No ground truth check

Labels are not yet compared with star ratings.

Conclusion and future work

What we built

- End-to-end pipeline from raw reviews to dashboard
- VADER sentiment plus aspect detection
- Enterprise data layer for books and reviews
- Planned and tracked with ITPM tools

Next steps

- Aspect-based sentiment analysis (ABSA)
- Transformer models such as BERT
- Evaluate with precision, recall and F1
- Skip heavy cleaning before VADER